



भारतीय प्रौद्योगिकी संस्थान पटना
INDIAN INSTITUTE OF TECHNOLOGY PATNA
Bihta, Patna - 801106, Bihar, India
Department of Computer Science & Engineering

Head of the Department

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Certificate of Internship

This is to certify that Krish Sangal (CS-2341002), a student of IILM University, Greater Noida, successfully completed a Research Internship at the Indian Institute of Technology Patna (IIT Patna) under my supervision.

Internship Duration: June 1st, 2026 to 20th July, 2026

Project Title: A Causal Reinforcement Learning Framework for Informer Guided Auto-Scaling in Container Cloud Platforms.

During the internship period, Krish Sangal demonstrated exceptional dedication, commitment, and technical proficiency while working on the project of "A Causal Reinforcement Learning Framework for Informer Guided Auto-Scaling in Container Cloud Platforms." Under my guidance, he actively contributed to research activities, result analysis, algorithm development, and implementation. His work involved integrating an Informer-based load forecaster with a causal discovery layer, implementing a Soft Actor-Critic agent for elastic scaling decisions, and evaluating the framework against Kubernetes HPA and threshold-based predictive baselines on the 1998 World Cup and Alibaba cluster workloads within the AutoScaleSim and CloudSim Plus simulation environment.

Through his rigorous efforts and innovative approaches, Krish Sangal significantly enriched the project, making valuable contributions to its successful completion. His work has demonstrated his understanding of the subject matter and his ability to apply theoretical knowledge to real-world problems.

This certificate attests to Krish Sangal's participation in the internship program, showcasing his exemplary performance and enthusiasm for academic and research pursuits. I believe that he possesses immense potential and will continue to excel in his future endeavors.

We extend our best wishes for his continued success and growth in the field of Computer Science and Engineering.

Prof. Rajiv Misra
(HoD, CSE)

